MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, HI-TORQUE, TYPE II

The complete requirements for procuring the screwdriver bits described herein shall consist of this document and the latest issue of Specification MIL-B-9946.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The tool applicable herein shall be a screwdriver bit assembly for use with recessed drive screws and bolts conforming to NAS3375O.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

QQ-T-570	Tool Steel, Alloy
GGG-W-641	Wrench, Socket; (and sockets, handles and attachments for socket wrenches; hand)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Single copies of Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LOMS, Federal Center, Battle Creek, Mich. 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5120

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STANDARDS

Military

MS33750	Recess-Hi-Torque, Dimensions of Recess Gage, and Driver for

National

NAS 583 thru 590	Bolt, 100° Flush Tension Head, "Hi-Torque" Recess, Alloy Steel, 160,000 PSI.
NAS 1243 thru 1250	Bolt, 100° Flush Tension Head, "Hi-Torque" Recess, .0156 Oversize Shank, Short Thread, Alloy Steel, 160,000 PSI

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) INC., STANDARDS:

B107.4 Driving and Spindle Ends for Portable Hand, Air and Electric Tools

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Design. The tool shall be an assembly of a square female socket part, with a square socket conforming to ANSI B107.4 and GGG-W-641, with an extension on the opposite end from the socket fitted with a circular wheel shaped part for driving screws. For further reference the socket equipped part will be referred to as the "adapter" and the wheel shaped part will be the "driving blade". This driving blade shall be of the required configuration for driving screw-head fasteners as applicable to MS33750. The assembly shall have provisions for adjusting the driving blade to any peripheral position and for snugly securing it in place in any adjusted position.

3.2 Material.

a. SAE 4140 (or equal) for adapter.

b. QQ-T-570, Type S2 (or equal) for driving blade.

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3.3 Hardness. 58 to 64, Rockwell "C" scale.

3.4 Coating. Rust preventive, for interim storage.

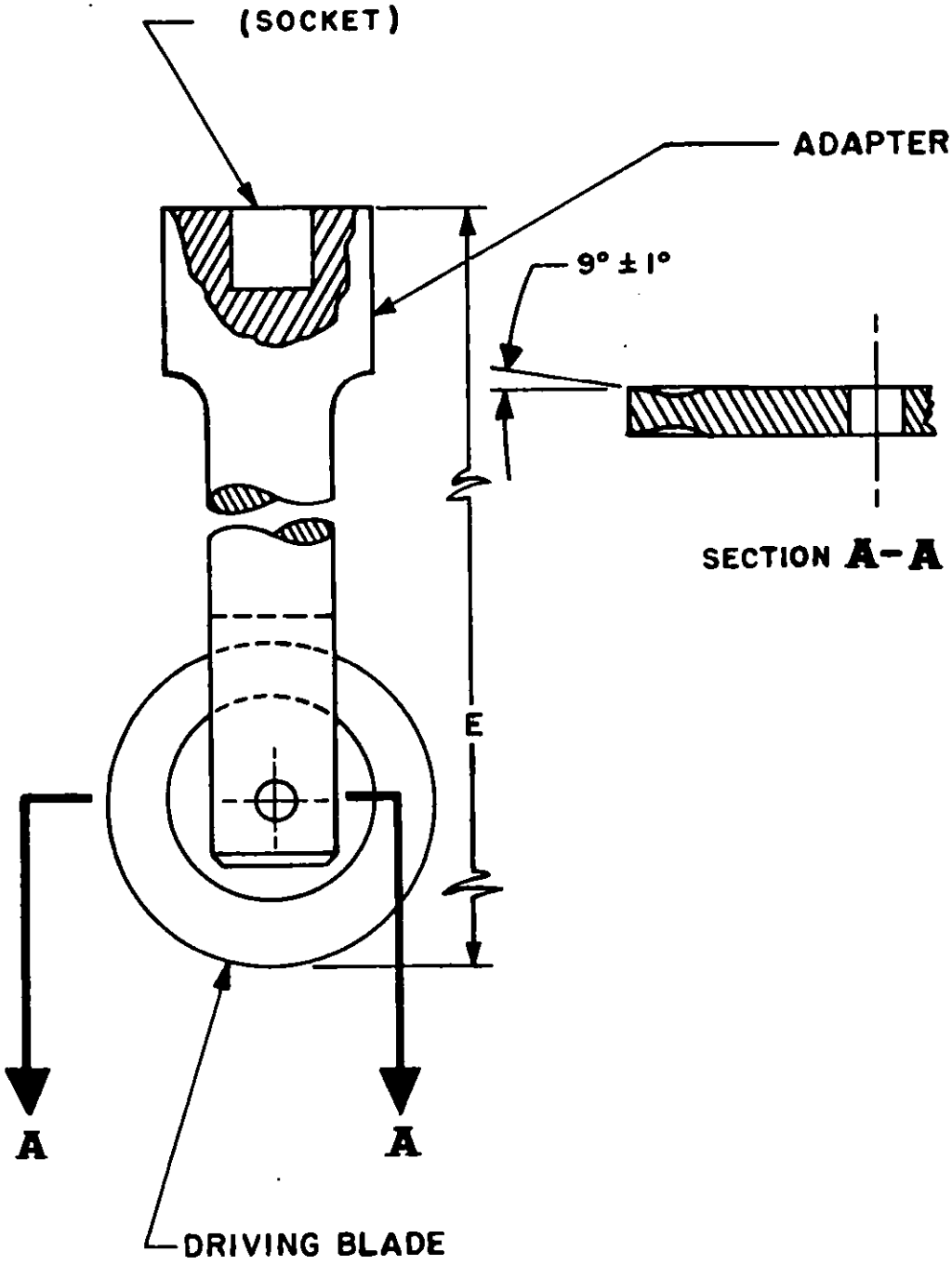
3.5 Configuration. Details of configuration as shown on Figure 1, Table I, and as applicable for usage on screw and bolt heads in accordance with MS33750 shall be complied with.

TABLE I - DETAILS OF TOOL ASSEMBLY

DRIVER NO SIZE (Ref: MS33750)	TOOLS APPLICABLE TO THIS SPECIFICATION SHEET			
	DRIVING BLADE DESIGNATION	TOOL ASSEMBLY		
		Designation	Socket Nom. Size (in.)	Nom. Length (inches)
HTS-3	HTB3M	HT2303M	1/4	1-63/64
HTS-4	HTB4M	HT3304M	3/8	2-43/64
HTS-5	HTB5M	HT3305M	3/8	2-47/64
HTS-6	HTB6M	HT3306M	3/8	2-47/64
HTS-6	HTB6M	HT3406M	1/2	3-11/64
HTS-7	HTB7M	HT3407M	1/2	3-15/64
HTS-8	HTB8M	HT3408M	1/2	3-19/64
HTS-9	HTB9M	HT3409M	1/2	3-27/64
HTS-10	HTB10M	HT3410M	1/2	3-35/64

NOTE: Dimensional details, as applicable, are shown on MS33750.

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TOOL ASSEMBLY

FIGURE 1

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4. QUALITY ASSURANCE PROVISIONS

4.1 Test Block. The test block to be used for sample testing shall be manufactured by the supplier with a testing slot in accordance with the applicable screw driving slots as shown on NAS 583 through NAS 590 or NAS 1243 through 1250 with the exception that the testing slot shall accommodate from 57 to 62 degrees angular portion of the driving blade being tested. A comparison of driving blades to applicable NAS numbers is shown in Table II.

4.2 Tool testing. Sample testing of driving blades shall withstand torque forces shown in Table II without any indication of failure.

TABLE II - NAS COMPARISON AND TORQUE TESTING REQUIREMENTS

Blade Designation	Compatible NAS Bolts	Min Fatigue Torque Test Load (Inch-Pound)	Min Ultimate Torque Test Load (Inch-Pounds)
HTB3M	583 or 1243	70	135
HTB4M	584 or 1244	170	340
HTB5M	585 or 1245	340	675
HTB6M	586 or 1246	500	1000
HTB7M	587 or 1247	900	1800
HTB8M	588 or 1248	1350	2700
HTB9M	589 or 1249	1900	3800
HTB10M	590 or 1250	2700	5400

5. PREPARATION FOR DELIVERY

(This section is not applicable. It is covered in MIL-B-9946.)

6. NOTES

6.1 Ordering Data. The applicable item required should be identified as follows: "MIL-B-9946, Type II, Part number HT---" (reference column 2 or 3 of Table I for the part number required).

Custodians:
Army - GL
Navy - AS
Air Force - 99

Reviewers:
Army - AV, MI, MC
DSA - CS
Air Force - 11

Preparing Activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-9946/2 Bit, Screwdriver, Hi-Torque, Type II

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

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